

THE NEXT AI IMPERATIVE – SERIES

Insights on Power, Principle, and Stewardship in the Intelligent Age

The Next AI Imperative

Governing Intelligence at Scale

Andreea Bulisache & Dr. Fumbi Chima

A Strategic Framework
for Boards, Investors, and Policy Leaders Navigating the AI Economy

October 2025

The Governance Decade

The Decade of Intelligent Governance: *Why This Series Exists.*

The coming decade will determine whether artificial intelligence becomes humanity's greatest multiplier or its most complex liability. Between 2025 and 2035, leaders will be tested not on how they deploy intelligence, but on how they govern it.

This anthology, **The Next AI Imperative**, captures that inflection, an evolving blueprint for boards, investors, and policymakers navigating intelligence as infrastructure.

Across these article, we explore how capacity, geopolitics, sustainability, ethics, and capital now converge into a single leadership frontier: governing value at scale.

This is not just a decade of innovation. It is the decade of intelligent governance.

The Next AI Imperative

Co-authored by Andreea Bulisache and Dr. Fumbi Chima, **The Next AI Imperative** series was born out of a shared conviction: that leadership in the age of intelligent systems demands governance, not guesswork.

What began as a set of articles on capacity, geopolitics, sustainability, and ethics has evolved into a live framework for boards and investors navigating AI's systemic transformation.

Future editions will expand into global dialogues, sector analyses, and a forthcoming book dedicated to governing intelligence at scale.

Foreword

A Framework for Governing Intelligence

Reframing the AI Conversation

A New Era of Leadership

Artificial Intelligence has evolved from disruptor to organizing principle — a structural force reshaping how nations assert power, how enterprises create value, and how boards govern complexity.

What began as a race for innovation is now a test of **governance, foresight, and systemic intelligence**.

AI flows through every artery of the economy — compute, energy, capital, and policy — demanding leaders who think in **systems, value chains, and ecosystems**.

The question is no longer how fast we deploy AI, but how wisely we govern it — ensuring that value is created, retained, and sustained.

AI is no longer an instrument of innovation; it is the new architecture of value, power, and responsibility.

This collection, *The Next AI Imperative*, brings together six essays written at a pivotal inflection point in AI's evolution.

Each explores a distinct dimension of this shift — from capacity and geopolitics to sustainability, ethics, and capital — and together they form a single thesis:

Governing intelligence at scale is the defining leadership challenge of our time.

The Authors

Architects of Value and Systems Thinkers in Practice



Andreea Bulisache

Andreea Bulisache is a global strategist and board advisor shaping how organizations govern intelligence at scale. A former Microsoft executive, she has led international programs in AI, cybersecurity, and data governance — translating emerging technologies into enterprise value and investor confidence.

As Founder of Stratified Advisory and Advisor with Cantellus Group, Andreea works with boards, investors, and CEOs to embed governance into innovation — ensuring that AI-driven growth remains resilient, ethical, and value-accretive.

A graduate of Harvard Business School's Women on Boards Program, she combines systems thinking, regulatory foresight, and capital discipline to guide organizations at the intersection of technology, policy, and capital. Her work focuses on how boards can steward AI responsibly — balancing opportunity, risk, and sustainable scale.



Dr. Fumbi Chima

Dr. Fumbi Chima is a globally respected technology executive and board director with over two decades of leadership across industries — including adidas, Burberry, Walmart, Boeing Credit Union, and Fox Networks.

Her expertise lies in transforming large-scale operations through digital strategy, cybersecurity, and responsible innovation — aligning technology with measurable business outcomes and cultural integrity.

Dr. Chima's leadership is grounded in value stewardship: the disciplined alignment of innovation, ethics, and long-term enterprise resilience. Her board roles and global executive experience give her a panoramic perspective of the ecosystems where business, policy, and technology converge — making her a sought-after voice on the governance of AI-enabled transformation.

Together, Bulisache and Chima embody a rare dual fluency — uniting strategic foresight with operational mastery, board governance with policy insight, and innovation with accountability. Their shared ethos: technology must serve human systems, not destabilize them.

The Arc of Inquiry

The six articles in this series are not discrete commentaries; they form an integrated framework — an evolving systems map of how AI reshapes governance, value, and resilience.

1. The Next AI Imperative: Capacity, Geopolitics, and Sustainability

Introduces the trilogy of forces driving systemic disruption — capacity, policy, and sustainability — and reframes AI as the connective tissue of the modern economy.

2. Capacity: From Model to Megawatt - Rethinking AI Leadership Through Infrastructure

Exposes the new economics of intelligence, where compute and energy capacity define both innovation velocity and strategic constraint.

3. Geopolitics and the AI Power Play

Explores how nations and corporations are redrawing alliances through digital infrastructure and data sovereignty — turning computation into capital and strategy.

4. Sustainability at Scale - Rethinking AI's Energy, Ethics, and Governance

Addresses the paradox of progress: the same AI systems that can accelerate sustainability may also undermine planetary stability if not governed with systemic accountability.

5. Ethics, Sovereignty, and Cyber-Resilient Systems

Argues that resilience and ethics must be engineered into architecture — where sovereignty is no longer national, but digital.

6. Private Equity's AI Infrastructure Play: From Megawatts to Multiples

Examines how investors are funding the backbone of the AI economy and why value retention depends on disciplined stewardship, not speculative acceleration.

A Mirror and a Mandate

Across these articles, a unifying logic takes shape:

- Capacity is the new currency of innovation.
- Sustainability is the new measure of legitimacy.
- Geopolitics is the new context of competition.
- Ethics is the new infrastructure of trust.

Together, these dimensions form both a mirror and a mandate — a mirror reflecting the systemic interdependencies shaping this era, and a mandate for boards and leaders to design with systems intelligence — balancing innovation with governance, and ambition with accountability.

The Road Ahead

In the following pages, you will find each article in its extended format, as originally published.

Individually, they explore distinct dimensions of the intelligent age — from capacity and geopolitics to sustainability, ethics, and capital.

Together, they form a system of systems — a map for understanding how intelligence transforms governance, value creation, and institutional resilience.

The series invites directors, policymakers, and investors to widen their horizon — from implementation to orchestration, from isolated projects to interconnected ecosystems, from short-term performance to enduring value stewardship.

In this new era, leadership will be measured not by speed, but by stewardship — by the ability to see the whole, connect its parts, and ensure that technology compounds value across generations.

The Next AI Imperative is not a framework, but a lens and a call to action — for those who understand that governing intelligence is not about control, but about shaping the architecture of sustainable value.

For collaborations, speaking engagements, or media inquiries:

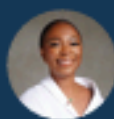
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The Next AI Imperative: Capacity, Geopolitics, and Sustainability

The questions boards should be asking



Fumbi Chima



Andreea Bulisache

This article is the first in a four-part series designed to reframe the AI conversation for senior executives and board directors.

By now, everyone can agree on the benefits of AI in the areas of increased productivity, faster workflows, and streamlined capabilities. Boardrooms and executives worldwide are excitedly discussing use cases, pilot programs, ethics, and enhancements, like the promise of generative models. The current discourse tends to center on the “how”—how to implement, how to integrate, how to scale, how to govern. But one question is being overlooked: What’s the long-term cost of AI—not only financially, but as it pertains to capacity and access, geopolitics, and sustainability?

CIOs and corporate directors are expected to be stewards of innovation and risk. Yet, we believe there’s a missing dimension to the conversation. AI touches everything from national security to climate resilience. As adoption accelerates, boards must grapple with three critical interconnected issues:



Capacity

AI’s non-stop demand for computing power and energy. Who controls access, and how does that affect resilience and scale?



Geopolitics

How do emerging global regulations, trade policies, and diplomatic tensions influence AI strategy and sovereignty?



Sustainability

Can businesses reconcile the carbon footprint of AI with their environmental commitments and values?



These questions should be on every boardroom agenda. Not expecting and planning for AI’s effects could lead to strategic misalignment, increased regulatory scrutiny, and long-term operational risk. This article serves as a guide for leaders as they navigate the fast-evolving AI landscape.

Capacity: The AI advantage no one is talking about

When we talk about AI readiness, the conversation often starts with talent, tools, and data. But there's a fundamental layer that shapes who gets to scale, and who gets left behind. *It's capacity.* AI demands massive computing power, which needs massive amounts of energy. Training a single large language model like GPT-4 consumes an estimated 1,287 MWh (the annual energy use of roughly 130 U.S. homes (MIT NEWS, 2024). Whoever has the energy has the upper hand. Capacity (access to energy and compute) may be the greatest competitive advantage.

The companies poised to win in AI are speeding to secure infrastructure. In 2023, [Microsoft pre-committed](#) to purchasing 485,000 NVIDIA H100 GPUs. This was their way of taking control over the energy needed to power their AI advances. The ability to source, afford, and scale *energy* access is inseparable from AI strategy. While tech giants in the U.S. make bold precommitments to secure AI capacity, other regions are racing to catch up. Take Europe, for example. Its disjointed energy grid and dependence on imported semiconductors leave it particularly vulnerable. The [EU Chips Act](#) and the [European Green Deal](#) aim to improve authority over areas, but whether Europe can move fast enough to stay in the game is yet to be determined.

This global gap has real implications for businesses. Companies without long-term compute strategies, or those overly reliant on third-party providers, may find themselves priced out or left behind. Adding even more complexity is the fact that the majority of the world's chips are created in Asia, particularly Taiwan and South Korea. U.S.-China tensions are rising. Rising tensions may impact availability, which in turn will impact capacity. Providers may be forced to prioritize capacity based on political and economic considerations, not just contracts. Boards must consider how geopolitical risks may restrict capacity, altering pricing and access.

Boards and CIOs should ask:

- ▶ What percentage of our AI workloads depend on external providers?
- ▶ Do we have a long-term strategy for securing energy and compute capacity?
- ▶ Are we vulnerable to geopolitical risks in the semiconductor supply chain?
- ▶ Should we build, buy, or partner to secure our AI infrastructure in the future?

Geopolitics and the ever-evolving AI regulatory complexity

Which leads us into the next concern: AI advancement is entangled within the global geopolitical landscape. The most notable tension currently lies between the U.S. and China. Recent U.S. export restrictions on advanced AI chips are intended to slow China's military and economic advancement as we have seen in the recent embargo of an additional dozen Chinese companies on the export blacklist. In response, China is accelerating domestic semiconductor production and setting up its own AI development ecosystem.

These moves lay the foundation for a potential "AI splinter net"—separate, regionally defined ecosystems governed by different values, policies, and technical standards. Companies working across borders may face increasing barriers to compute infrastructure, talent mobility, and cross-border data flow. What was once a global platform for innovation is becoming more fractured and politically influenced.

At the same time, AI governance frameworks are tightening across the world. According to consulting firm McKinsey, the EU's AI Act delivers a tiered approach to risk and compliance, while China promotes AI models that align with state-approved content and policies. These regulations affect how AI systems are trained, monitored, and maintained. In short: compliance just became even more complex, and AI strategy is inseparable from regulatory strategy. Boards must shift from reactive compliance to proactive planning and be prepared to vary their approach based on jurisdictions. Companies must be ready to adapt. Beyond governance, political pressure around automation and job displacement continues to build. In some areas, we may see regulations that intentionally slow or restrict AI-driven automation to preserve employment or align with domestic policy priorities. At the same time, some sustainability mandates push companies to account for AI's energy usage and environmental impact, further complicating deployment at scale.

Boards and CIOs should ask:

- ▶ How do regional AI regulations impact our global operations?
- ▶ What contingency plans exist if geopolitical shifts disrupt our AI supply chains?
- ▶ Are we aligning our AI strategy with data sovereignty and localization requirements?
- ▶ How are we balancing automation ambitions with emerging labor and ESG expectations?



Sustainability: AI's Hidden Environmental Costs

The conversation around AI's carbon footprint is still catching up to its technological momentum. While some boards have made progress with ESG and climate targets, many have yet to fully consider the climate implications of AI at scale. At the core of this issue is a concept known as **Jevons' Paradox**, which states that as efficiency improves, consumption often increases. Applied to AI, this means that even if algorithms become more energy-efficient, the sheer volume of AI usage and the business incentives behind it can still drive up overall energy demand.

For example, AI data centers need water to stay cool and avoid overheating, which may contradict some of AI's efficiency gains. On the other hand, AI may also be used positively in climate action, such as by analyzing satellite imagery or detecting deforestation. The balance sheet between these examples can become foggy. These effects are often invisible in standard AI ROI conversations but carry real consequences.



There is also a deeper structural challenge at play. The most prominent AI breakthroughs today—think large language models—reward growth and scale over sustainability. As long as market forces and short-term returns primarily drive innovation, aligning AI development with climate goals will remain difficult unless it also serves those same incentives.

But the climate stakes are non-negotiable. A recent report from MIT notes that scientific consensus warns that global warming must be limited to 1.5°C to avoid the worst-case climate change scenarios. Every sector, including technology, must adapt. Boards must consider whether their AI ambitions align with their sustainability commitments or whether they risk building systems today that will conflict with policy, investor expectations, or social license tomorrow.

The question is no longer whether AI has an environmental impact. It's how boards will choose to respond—and whether they can lead with clarity as the regulatory and public landscape evolves. Boards must urgently evaluate whether their AI ambitions align with sustainability commitments and consider potential conflicts with policy, investor expectations, and social accountability.

Boards and CIOs should ask:

- ▶ How are we measuring and managing the environmental impact of our AI initiatives?
- ▶ Are our AI growth plans aligned with our stated ESG goals and commitments?
- ▶ What actions are we taking to mitigate AI-driven increases in resource consumption?
- ▶ Have we assessed the long-term regulatory and reputational risks associated with our current AI sustainability practices?



How can boards move forward: Frameworks to consider

Boards and CIOs need robust frameworks to assess and guide their AI strategies through a lens that includes climate impact, regulatory readiness, and long-term risk. We recommend starting with a combination of quantitative and qualitative assessments. Companies should consider regularly stress-testing AI infrastructures against energy volatility and shifting regulations and conducting scenario planning exercises to anticipate geopolitical disruptions. Finally, they should evaluate integrating sustainability metrics into AI procurement and infrastructure decisions.

Ultimately, proactive boards will mitigate risks, align AI with core business values, and gain stakeholder trust. Responsible AI strategies position companies for sustainable, resilient growth in an increasingly scrutinized business landscape.

AI is not just the next wave of digital transformation; it's a fundamental shift in how businesses operate, compete, and contribute to the world. As leaders, we are only beginning to understand the true scope of what this means. The questions we ask today will shape tomorrow's AI ethics, infrastructure, and global impact. This is about fulfilling our duty to shareholders and our responsibility as global citizens.

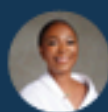


In our next article, we'll uncover the hidden battleground of AI capacity, revealing why control of computing infrastructure and energy resources will define strategic winners. Companies overlooking these invisible factors risk being left behind in AI's next evolution.



Capacity: From Model to Megawatt – Rethinking AI Leadership Through Infrastructure

The questions boards should be asking



Fumbi Chima



Andreea Bulisache

This article is the second in a four-part series designed to reframe the AI conversation for senior executives and board directors.

Capacity is the invisible force shaping AI's future. As leaders passionate about guiding enterprises and boards through pivotal moments, we believe that the capacity conversation — access to compute, energy, and resilient infrastructure — must move from the server room to the boardroom. This article continues our series launched with "[The Next Imperative](#)" framing why capacity is now a board-level priority and why tomorrow's market leaders will be those who understand that scaling AI is not merely a technical issue—it is a geopolitical, operational, and existential one.

Capacity as Strategic DNA

When AI strategies falter, it won't be for lack of vision, talent, or even capital. It will be because companies underestimated the race for capacity. Boards that frame AI as a "tech team issue" risk being left behind. As adoption accelerates, the ability to secure and manage capacity will become a critical pillar of enterprise resilience, competitiveness, and innovation.



Capacity must be seen as a living, breathing part of corporate strategy—no longer an afterthought, but a core competency. As organizations pivot to AI-driven models, the race to secure sustainable, scalable capacity will be one of the defining differentiators between those who thrive and those who falter.

In a world where AI training and deployment devour resources at unprecedented rates, capacity isn't a background concern—it's a currency. Every board must ask: **Who controls the compute, energy, and infrastructure we will need to thrive?**

(Source: International Energy Agency, 2024)

Training a next-generation AI model is already comparable to powering a small town for a year. The leap to multimodal, autonomous AI systems will only deepen this gap. Companies failing to plan for nonlinear scaling will find themselves frozen—outpaced by competitors who treated capacity as strategy, not logistics. (Source: RAND Corporation, 2024)

Moreover, control over compute is rapidly becoming the difference between market makers and market takers. Strategic planning must now account for capacity bottlenecks as rigorously as financial risks.

Control over AI infrastructure is rapidly consolidating. Technology giants are no longer just building platforms—they are securing the lifeblood of future innovation. When Microsoft pre-committed to nearly half a million GPUs, it wasn't just to train models faster. It was to create a strategic anchor (Source: Financial Times, 2024). In China, investments in domestic chip production signal that capacity is being viewed through a lens of national security, not merely corporate competition. Europe's push with initiatives like the EU Chips Act is a recognition that without domestic capacity, sovereign AI leadership is impossible. (Source: European Commission, 2023)

Companies must rethink alliances and dependencies. Geographic and political risks now play as large a role as technical leadership. There is an emerging premium on flexibility—the ability to shift compute loads across regions, jurisdictions, and providers as needed. (Source: Brookings Institution, 2024)

Boards and CIOs should ask:

- ▶ How exposed are we to capacity shortages or price shocks?
- ▶ Is our AI ambition matched by a roadmap to secure the energy and compute it requires?
- ▶ How are we tracking shifts in AI hardware and software requirements that may affect our capacity models?
- ▶ Are we comfortable with our reliance on hyperscalers, or should we diversify our capacity sources?
- ▶ Do we have influence or access strategies in place if geopolitical shifts restrict global compute markets?
- ▶ What regulatory trends might alter our current access to global infrastructure?

The infrastructure bottleneck is becoming critical. Securing rack space, cooling systems, and energy isn't a "nice to have"—it's now a determinant of who can scale AI and who can't. Electric grids are already under strain. In many regions, lead times for new connections now exceed five to seven years. Companies assuming the cloud will endlessly scale to meet demand without local infrastructure constraints are betting against physics, policy, and climate volatility. (Source: IEA, 2024; CSIS, 2024)

Innovation in grid flexibility, green energy integration, and data center design is uneven globally. Leaders must prioritize infrastructure intelligence—proactive, real-time tracking of regional risks—as part of operational resilience.

Questions boards should be asking about sustainability:

- ▶ Are our AI growth plans stress-tested against regional infrastructure constraints?
- ▶ Should we engage proactively in grid modernization initiatives or form energy procurement partnerships?
- ▶ What is our contingency plan if primary cloud partners face infrastructure bottlenecks?

Every AI milestone we celebrate has a carbon, water, and resource footprint. The myth that digital progress is weightless is being shattered. Energy and water consumption for AI data centers is projected to rival that of entire nations. Worse, the pace of improvement in efficiency is not keeping up with the exponential scaling of AI deployment. (Source: Deloitte, 2025)

Boards that treat sustainability and AI as separate discussions are missing the point: the two are now inseparable. Investors, regulators, and the public are beginning to scrutinize the environmental footprint of AI with the same intensity as traditional industries. New ESG reporting frameworks will likely require explicit accounting for AI-driven resource consumption.



Questions boards should be asking about sustainability:

- ▶ Are our ESG disclosures accounting for AI-driven energy and water impacts?
- ▶ How do our AI ambitions align—or conflict—with our public sustainability commitments?
- ▶ Are we considering innovations in "green AI" practices to mitigate environmental costs?

Boards must stop viewing capacity as an operational line item. The forward-leaning organizations will treat it as a strategic differentiator. Innovators are already moving:



Forming direct renewable energy procurement deals tied to AI expansion.



Building regional micro-data centers to sidestep national infrastructure chokepoints.



Engineering AI workflows that optimize for "green compute" windows where renewable energy is abundant.



Prioritizing hardware and software efficiency in model selection to minimize unnecessary compute usage.



Incorporating energy-aware AI algorithms and adaptive scheduling models that align workload peaks with renewable energy availability.

Capacity-Conscious Strategies for Boards:

- ▶ **Shift AI Workloads to Green Data Centers:** Direct compute tasks to renewable-rich regions through dynamic workload orchestration.
- ▶ **Optimize AI Models:** Implement techniques like quantization, pruning, and knowledge distillation to reduce compute needs.
- ▶ **Adopt Low-Power Chips:** Use specialized hardware like NVIDIA H100, Google TPU v5e, AWS Inferentia to boost efficiency.
- ▶ **Pursue Policy and Regulatory Engagement:** Proactively shape and comply with emerging standards for sustainable AI growth.

Capacity resilience is fast becoming a proxy for enterprise agility. Organizations that fail to embed capacity thinking into their AI strategy will struggle with regulatory compliance, stakeholder trust, and operational scalability.

The New Board Agenda: Leading with Foresight

What we advocate is simple, but urgent: Boards must integrate capacity resilience into the heart of enterprise risk management, innovation planning, and sustainability strategy.

- ▶ **Scenario Planning:** Model AI capacity needs under multiple geopolitical and regulatory futures.
- ▶ **Investment Shifts:** View investments in compute, energy resilience, and sovereign cloud capabilities as long-term strategic assets.
- ▶ **Cross-Functional Governance:** Embed AI capacity considerations into sustainability, security, and M&A committee mandates.
- ▶ **Talent Strategy:** Ensure technical leadership roles (CIO, CTO, Chief Sustainability Officer) are collaborating to address capacity resilience cohesively.
- ▶ **Stakeholder Communication:** Transparently articulate capacity strategies to investors, regulators, and customers as a mark of enterprise responsibility.
- ▶ **Risk Frameworks:** Expand traditional enterprise risk frameworks to incorporate scenarios such as prolonged supply chain interruptions, energy price volatility, and regulatory shocks.

Boards that act today—building foresight, resilience, and influence into their AI capacity strategies—will not only enable innovation. They will safeguard competitiveness, stakeholder trust, and enterprise value.



Conclusion: Capacity Will Define Strategic Winners

AI will not merely reward those who innovate fastest. It will reward those who prepare most wisely. The next wave of market leaders will be those who recognized that capacity is not infrastructure—it is destiny. Capacity is the ultimate forcing function for strategic clarity. It demands organizations confront uncomfortable truths about scale, resilience, sovereignty, and sustainability.

Boards that engage with this challenge—with urgency, creativity, and courage—will not only shape their organizations' futures but will help shape the societal impact of AI itself.



In our next article, we will explore the shifting geopolitical landscape where capacity, alliances, and sovereignty will decide the future of AI leadership.



The Next Imperative: Geopolitics and the AI Power Play

The questions boards should be asking



Fumbi Chima



Andreea Bulisache

This article is the third in a four-part series designed to reframe the AI conversation for senior executives and board directors.

Artificial Intelligence (AI) has moved beyond its origins as a research endeavor to become a central pillar of global strategic competition. Following *The Next Imperative and Capacity: From Model to Megawatt*, we now turn to the shifting geopolitical landscape, where AI is redrawing lines of influence, sovereignty, and security.

Across economies and institutions, AI is increasingly recognized as more than a technology—it is a lever of national security, economic development, and ideological governance. The race for AI leadership now encompasses not only algorithms, but also compute capacity, access to energy, talent mobility, and influence over regulatory frameworks. This convergence signals a shift that boards can no longer afford to view from the margins.

Drawing on our advisory work with organizations navigating AI strategy, policy uncertainty, and systemic transformation, we approach this topic with a dual lens: **one grounded in geopolitical analysis, and the other in enterprise foresight.** We write from the conviction that governance must evolve alongside innovation—and that boards must be equipped not only to adapt, but to shape the environment in which AI operates.



Boards today are called to engage with AI not only as a growth enabler but as a geopolitical force—one that affects operating models, capital flows, supply chain dependencies, and stakeholder trust. Decisions made at the highest levels of governance will shape how enterprises navigate global complexity and contribute to the long-term security and sustainability of the ecosystems in which they operate.

Organizations that embed geopolitical foresight into their AI strategies will strengthen their agility and capacity to lead in a world of accelerating realignment. Those that do not may face rising exposure—legal, operational, and reputational—in markets where AI is both a source of power and a strategic lever.

AI as a New Instrument of Sovereignty

Technological revolutions have long influenced the evolution of global power structures. AI is now extending this legacy—transforming how countries pursue strategic advantage, protect infrastructure, and influence the global digital order.

Governments around the world are integrating AI into national security doctrines, economic development plans, and digital governance models. The objective is not only to leverage AI, but to control its foundations—compute infrastructure, data flows, semiconductors, and energy.

In this context, AI systems are increasingly treated as sovereign-grade assets. Their development and deployment are shaped by export controls, investment restrictions, and national capability strategies. For globally active enterprises, this creates a new strategic landscape—one where navigating political priorities is as essential as managing platform roadmaps or customer pipelines.

AI's influence extends across domains:



Military modernization
strategies



Cyber defense
architectures



Next-generation industrial
competitiveness

Nations that achieve leadership in AI capabilities are likely to gain disproportionate leverage across trade, finance, and digital ecosystems. (Source: Brookings Institution, 2024; WEF Global Risks Report, 2024)

Boards and CIOs should ask:

- ▶ How do emerging views of AI as a sovereign asset affect our global operating models?
- ▶ Are we adequately monitoring how national AI strategies could influence market access, partnership dynamics, or regulatory exposure?

Strategic Shifts: Fragmentation, Infrastructure, and Regulation

The early vision of a seamlessly interconnected digital economy is giving way to a more fragmented reality, accelerated by AI's strategic significance. Distinct regulatory, technological, and geopolitical blocs are emerging:

- ▶ **The U.S.- led Alliance** focuses on private-sector-driven innovation in foundational models, hyperscale infrastructure, and trusted AI systems.
- ▶ **China's Sovereign Strategy** prioritizes domestic semiconductor production, national R&D investments, and active leadership in setting global AI norms. (Source: Reuters, 2024)
- ▶ **The EU and Responsible AI Bloc** advances ethical frameworks emphasizing risk management and human rights, balancing innovation with regulatory safeguards.
- ▶ **BRICS and Emerging Powers** are forging regional collaborations aimed at reducing dependency on dominant Western and Chinese technologies. (Source: Foreign Policy, 2024)



Beneath these strategic narratives lies a critical competition over control of infrastructure:

- ▶ **Semiconductors** are now viewed as strategic commodities, with export controls reshaping global supply chains. (Source: European Commission, 2023)
- ▶ **Data sovereignty** regulations require enterprises to localize storage and processing across jurisdictions.
- ▶ **Energy availability**, particularly access to low-carbon, high-reliability power sources, is becoming a central factor in AI scalability. (Source: IEA, 2024)

Diverging regulatory environments further amplify complexity:



The **EU AI Act** introduces binding obligations for high-risk AI systems, emphasizing transparency and accountability.



U.S. executive orders on AI governance prioritize national security and trustworthy AI principles.

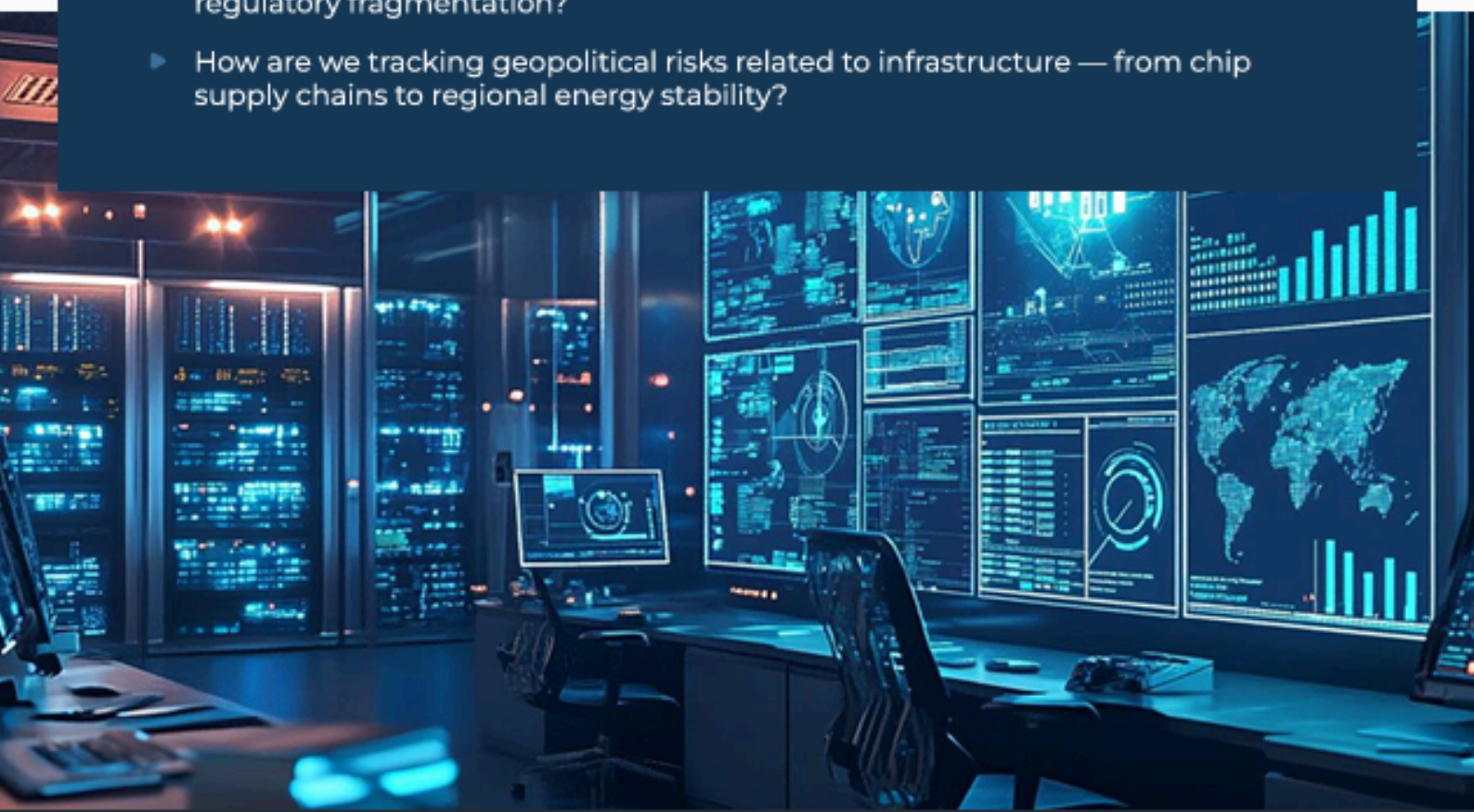


China's evolving AI regulations balance acceleration with censorship and information control. (Source: OECD AI Governance Briefs, 2024)

Across these dimensions, governments are deploying sovereign AI strategies as instruments of long-term economic leadership.

Boards and CIOs should ask:

- ▶ Are our data governance, cloud, and AI model deployment strategies resilient to regulatory fragmentation?
- ▶ How are we tracking geopolitical risks related to infrastructure — from chip supply chains to regional energy stability?



Cybersecurity and Geopolitical Risk: Boardroom Imperatives

AI's integration into critical systems—from infrastructure and healthcare to finance and logistics—brings new dimensions to enterprise vulnerability. Where threat actors once targeted networks, they are now probing algorithms, training pipelines, and the datasets that underpin decision-making.

The spectrum of risk is expanding:



Model inversion attacks, data poisoning, and synthetic content generation introduce novel vulnerabilities.



Non-state actors, empowered by generative tools, are able to scale misinformation, deepen cyberattacks, and automate influence operations.



Sector-wide risk amplification emerges when AI systems coordinate across supply chains or core infrastructure. (Source: Chatham House Reports, 2024)

In this environment, cybersecurity is no longer a technical function—it is a pillar of strategic oversight. AI-specific vulnerabilities must be governed with the same discipline as financial exposure or compliance risk.

Boards and CIOs should ask:

- ▶ Have we integrated AI-specific cybersecurity risks into our enterprise-wide risk frameworks?
- ▶ Are we building internal capabilities to detect, respond to, and recover from AI-enabled threats?

Strategic Imperatives for Boards: Navigating the New Chessboard

In a geopolitical climate shaped increasingly by AI, enterprise success will hinge not only on technological investment, but on the depth of insight guiding governance. Boards have a critical role to play in setting the tone and structure for strategic foresight, ethical adaptation, and resilient infrastructure.



Scenario Planning and Strategic Foresight

Boards should mandate the development of AI-specific geopolitical scenarios, integrated into risk, compliance, and capital allocation frameworks. These scenarios should model disruptions such as chip embargoes, regulatory fragmentation, or regional AI bans—and provide action-oriented recommendations for adaptation.



Infrastructure Diversification and Compute Agility

Diversifying access to compute—geographically and across vendors—offers both resilience and strategic leverage. Sovereign cloud partnerships, modular data center strategies, and compute resource pooling can provide insulation from shocks while supporting long-term scalability.



Cybersecurity Integration and Model Integrity

Boards should ensure that AI model governance, threat monitoring, and adversarial testing are embedded within existing cybersecurity programs. Oversight should involve collaboration across CISOs, Chief AI Officers, and legal risk teams, with attention to third-party platforms and generative content risk.



Policy Engagement and Thought Leadership

Board support for executive participation in multilateral AI coalitions, standards bodies, and ethics councils strengthens institutional influence. Active engagement with policy trends enables enterprises to anticipate shifts—and contribute to the formation of principles that shape future regulation.



Board and Executive Capability Development

Geopolitically informed AI governance demands new fluency across the board. Boards should invest in director education that spans technology, digital governance, global compliance, and security. Cross-functional literacy among board members is a marker of maturity in AI oversight.



Supply Chain Mapping and Resilience Strategy

Organizations should maintain updated, granular maps of their critical AI-related supply chains—from semiconductors to energy infrastructure—and develop options for redundancy, friendshoring, and crisis response. Transparency and agility will define enterprise preparedness in increasingly contested environments.

Boards and CIOs should ask:

- ▶ How resilient are our AI ambitions against potential geopolitical fractures?
- ▶ Are we equipping leadership with the capabilities to anticipate, navigate, and influence emerging AI governance landscapes?

Conclusion: Geopolitics as a Central Pillar of AI Strategy

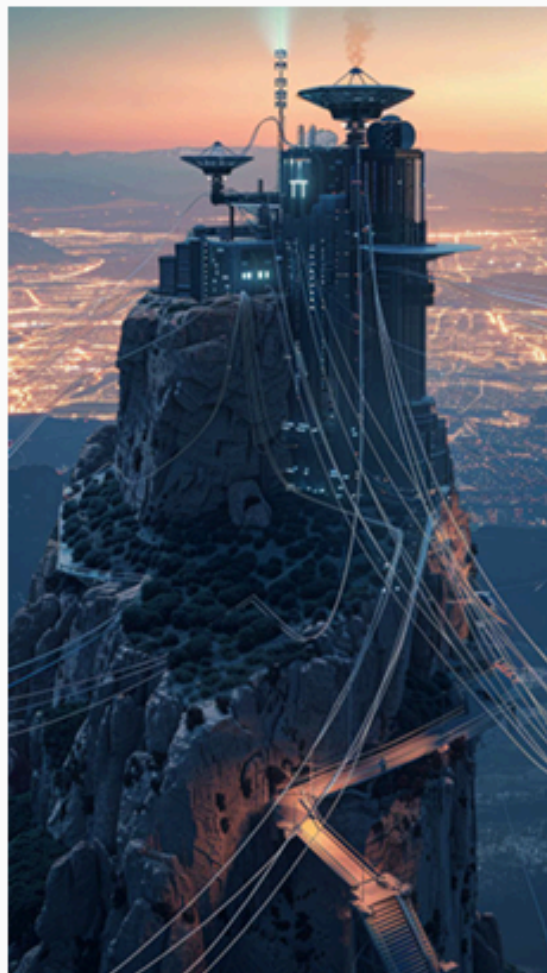
Artificial Intelligence is no longer merely a driver of digital transformation. It is a defining force in how nations assert influence, how economies compete, and how enterprises navigate opportunity and risk. Boards that frame AI through a narrow lens—focused only on ROI or product acceleration—will miss the broader implications of geopolitical entanglement. In contrast, boards that engage deeply with the evolving intersections of policy, power, and platform will position their organizations for durable leadership.

Risks of Neglecting Geopolitical Dimensions

A lack of engagement with AI's geopolitical context is not a neutral stance. It carries risk:

- ▶ **Operational Inefficiencies:** Unanticipated restrictions on data flows, compute access, or infrastructure partnerships can disrupt global operations.
- ▶ **Regulatory Penalties:** Organizations that lag in aligning with national and regional AI frameworks may face compliance violations or delayed market access.
- ▶ **Loss of Trust:** Stakeholders increasingly expect companies to demonstrate awareness of ethical and geopolitical concerns. Inattention can erode credibility with investors, regulators, and civil society.

Boards that lead with discipline and insight—shaping resilient AI strategies in light of global dynamics—will define a new standard of preparedness and principle. The coming decade will be shaped not solely by the pace of technical innovation, but by the strategic acumen with which organizations anticipate shifts, mitigate vulnerabilities, and lead responsibly.

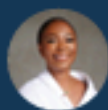


In our next and final installment, we will explore how sustainability considerations—from energy use to environmental impact—are rapidly becoming defining factors for responsible AI leadership.



The Next Imperative: Sustainability at Scale — Rethinking AI's Energy, Ethics, and Governance

The questions boards should be asking



Fumbi Chima



Andreea Bulisache

This article is the final in a four-part series designed to reframe the AI conversation for senior executives and board directors.

Artificial intelligence (AI) has moved from an emerging technology to a central strategic asset reshaping global industries. Following discussions around strategic foresight, computational capacity, and geopolitical considerations, we now focus on sustainability—an equally pivotal issue defining AI governance and leadership. The rapid deployment of AI across industries has created substantial environmental, ethical, and governance challenges. AI's intensive resource consumption poses risks to ecological stability, corporate credibility, and compliance standards.

For boardrooms, proactively embedding sustainability within AI strategies is not only about risk management but also about establishing leadership in a rapidly evolving global marketplace.

The Expanding Footprint of AI: A Growing Imbalance

The expansion of AI capabilities has been accompanied by a corresponding rise in resource demands. Training a large-scale language model now consumes energy equivalent to that used by 100+ U.S. households annually (Stanford University, AI and Global Trends Report, 2025).

Projections suggest that without intervention, emissions from AI infrastructure could rival those of the airline industry within a decade (International Energy Agency, Data Centers and Electricity Demand, 2024). Water consumption for data center cooling is adding further strain to ecosystems, particularly in water-stressed regions (MIT, AI's Environmental Impact: Explained, 2025).

According to a study by Nature Communications, unchecked AI-related energy consumption could significantly impact global temperatures (Nature Communications, Energy and Climate Implications of Artificial Intelligence, 2023). These developments invite a more holistic view of AI's role within global sustainability frameworks. Innovation cannot be decoupled from its material footprint.

Boards and CIOs should ask:

- ▶ How are we quantifying the full environmental impact of our AI initiatives?
- ▶ Are we aligning AI development goals with regional and global sustainability considerations?

Regulatory Pressures and Societal Expectations: A Tightening Net

Global regulatory frameworks are evolving rapidly in response to AI's environmental implications. The European Union's Green Deal and Chips Act embed sustainability criteria directly into technology funding initiatives (European Commission, EU Chips Act and Green Deal Initiatives, 2023). California's Climate Disclosure Laws require granular emissions reporting, encompassing digital infrastructure and cloud operations. Emerging regulatory initiatives in Asia and Latin America are introducing renewable energy sourcing mandates for future data centers (Asian Development Bank, Asia's Data Center Energy Challenge, 2023).

Simultaneously, stakeholder expectations are intensifying:



Consumers demand transparency regarding environmental impacts.



Institutional investors increasingly incorporate sustainability metrics into valuation models.



Talent pools — particularly emerging leaders — seek employers whose climate actions align with their values.

Boards and CIOs should ask:

- ▶ How prepared are we to navigate the next wave of environmental disclosure requirements?
- ▶ Are we positioned to lead in sustainability reporting, or will we be compelled to react under external pressure?

Rethinking Operational Strategy: Beyond Compliance, Toward Innovation

Leading organizations are reframing sustainability from a regulatory obligation into a driver of innovation and competitive advantage. Dynamic AI workload orchestration based on real-time grid carbon intensity is becoming a norm. Prioritization of energy-efficient model architectures through techniques such as pruning, quantization, and knowledge distillation can reduce compute needs by up to 70% (Accenture Labs & IIT Kharagpur, Towards Sustainable NLP: Insights from Benchmarking Inference Energy in LLMs, 2025).

Adoption of low-power, AI-optimized hardware to reduce both compute costs and emissions is accelerating. Importantly, AI is not only a contributor to energy consumption—it is becoming an enabler of sustainable energy management itself.



The AI-in-energy market is projected to expand from \$8.91 billion in 2024 to over \$58.66 billion by 2030, led by AI-driven predictive maintenance and smart grid optimization (GlobeNewswire, AI in Energy Market Projected to Skyrocket to \$58.66 Billion by 2030, 2024).

Boards and CIOs should ask:

- ▶ How are we embedding sustainability-driven innovation into our AI development and operational strategies?
- ▶ Are we adequately resourcing capabilities to optimize for both performance and environmental stewardship?

Expanding the Concept of Responsible AI

The evolution of Responsible AI frameworks invites a broader lens. Environmental responsibility must stand alongside transparency, fairness, and accountability as a core pillar of ethical AI deployment. Leading organizations are beginning to incorporate lifecycle impact assessments into model development and infrastructure planning.

Procurement practices are increasingly aligned to sustainability standards, extending expectations to partners, suppliers, and cloud providers.

Boards and CIOs should ask:

- ▶ Is environmental impact formally integrated into our Responsible AI framework?
- ▶ Are we setting measurable goals — and reporting progress transparently — on AI-driven resource consumption?



Strategic Imperatives for Boards: Leading Through Foresight

Embedding sustainability into AI strategy is no longer optional; it is a strategic necessity. Foresight-oriented governance will distinguish organizations able to scale AI responsibly and resiliently.

Board-level imperatives include:



Integrating sustainability risks into AI scenario planning and enterprise risk frameworks.



Aligning innovation investment with low-energy AI architectures, green infrastructure, and ethical supply chain initiatives.



Building cross-sector collaborations to advance standardization, transparency, and accountability in sustainable AI practices.



Leading voluntary disclosures, setting a tone of proactivity rather than compliance-driven response.

Boards and CIOs should ask:

- ▶ Are we shaping the narrative of sustainable AI leadership, or waiting to be shaped by it?
- ▶ Are we equipping the enterprise to navigate an increasingly resource-constrained, climate-conscious economy?



AI as a Catalyst for Sustainable Transformation

AI's potential to accelerate sustainability extends far beyond its own footprint. When applied strategically, AI can drive decarbonization, resource optimization, and environmental monitoring at scale:

- ▶ **Clean Energy Optimization:** AI is revolutionizing clean energy by optimizing power generation, improving grid management, and reducing carbon emissions by up to 50% (IEA, Data Centers and Electricity Demand, 2024).
- ▶ **Supply Chain Transparency:** AI-powered analytics can trace the carbon footprint of products and suppliers, enabling more sustainable procurement decisions.
- ▶ **Circular Economy Enablement:** AI-driven predictive maintenance and smart manufacturing can extend asset lifespans, reduce waste, and minimize raw material use.
- ▶ **Climate Risk Modeling:** Machine learning models can analyze satellite imagery to detect deforestation, monitor methane leaks, and predict extreme weather events, supporting more resilient infrastructure planning (MIT, AI's Environmental Impact: Explained, 2025).

Boards should recognize that AI, when governed responsibly, is a critical lever for achieving broader ESG and climate goals.

The AI-Sustainability-Ethics Nexus

The interplay between AI, sustainability, and ethics is reshaping the leadership agenda. Bias in AI algorithms, data privacy concerns, and the potential for job displacement all intersect with environmental stewardship. Boards must foster a culture of transparency, continuous learning, and cross-functional collaboration to address these challenges holistically.



Governance: The Board's Role in Sustainable AI

The board's responsibilities for governing sustainability and AI are fourfold:



Educating: Ensure the entire board, not just single-issue directors, is upskilled on AI and sustainability risks and opportunities.



Integrating: Make sustainability and AI cross-cutting themes in all risk management and strategic decision-making processes, not siloed topics.



Overseeing: Establish clear metrics for holding management to account against the company's sustainability and AI ambitions, and review progress regularly.



Ratifying: Publicly champion and ratify decisions made by management regarding the company's sustainability and AI journey, demonstrating unified board support.

Boards and management must work collaboratively, integrating sustainability and AI into every committee's lens—whether audit, risk, compensation, or strategy.



Regulatory Complexity and Global Fragmentation

AI's sustainability challenge is compounded by regulatory fragmentation. Distinct regional approaches are emerging:

- ▶ **The EU:** The EU AI Act and Green Deal embed environmental criteria into AI funding and governance, requiring transparency and accountability for energy use and emissions (European Commission, EU Chips Act and Green Deal Initiatives, 2023).
- ▶ **The U.S.:** Lacks comprehensive AI sustainability regulation, relying on sectoral standards and voluntary disclosures.
- ▶ **Asia and the Global South:** Context-specific frameworks are emerging, focused on aligning AI development with local social and environmental goals, and incentivizing sustainable innovation (Asian Development Bank, Asia's Data Center Energy Challenge, 2023).

Boards must anticipate and adapt to this evolving landscape, integrating scenario planning and proactive policy engagement into their governance frameworks.

The Paradox of AI for Sustainability

AI's promise as a tool for sustainability is paradoxically threatened by its own environmental cost. The energy and water demands of AI infrastructure risk offsetting the very gains it enables in other sectors (MIT, AI's Environmental Impact: Explained, 2025). This paradox demands a dual approach:

- ▶ Mitigate AI's own footprint through efficiency, green infrastructure, and responsible procurement.
- ▶ Maximize AI's positive impact by prioritizing applications that accelerate decarbonization, circularity, and resilience.





Operationalizing Sustainable AI: Strategic Imperatives for Boards

Integrate Sustainability into AI Governance

- ▶ Embed environmental KPIs into model approval frameworks, procurement, and M&A due diligence.
- ▶ Ensure that ESG disclosures explicitly account for AI-driven energy, water, and resource impacts.

Lead Transparent Disclosures

- ▶ Publish AI Sustainability Scorecards tracking energy efficiency, water reuse, and e-waste recovery.
- ▶ Set measurable goals and report progress transparently, positioning the enterprise as a leader in responsible AI.

Expand Responsible AI Frameworks

- ▶ Incorporate environmental responsibility as a core pillar of ethical AI, alongside transparency, fairness, and accountability.
- ▶ Extend sustainability expectations to partners, suppliers, and cloud providers.

Reimagine Partnerships and Infrastructure

- ▶ Co-invest with energy providers in green data corridors and grid resilience initiatives.
- ▶ Pursue sovereign cloud and regional alliances to align with local sustainability mandates and reduce exposure to energy volatility.

Future-Proof Talent and Culture

- ▶ Reskill teams in green AI engineering, lifecycle impact modeling, and sustainability analytics.
- ▶ Foster a culture where sustainability is woven into leadership incentives and innovation processes.

Scenario Planning and Foresight

- ▶ Integrate sustainability risks into AI scenario planning and enterprise risk frameworks.
- ▶ Model potential disruptions—such as energy price shocks, water scarcity, or regulatory tightening—and develop action-oriented responses.

The Path Forward: Balancing Innovation and Responsibility

AI is a powerful tool in the clean energy transition and broader sustainability agenda. But its unchecked deployment risks undermining its own promise. The path forward requires:

- ▶ Strategic investment in energy-efficient AI and green infrastructure.
- ▶ Robust regulatory frameworks that incentivize sustainable innovation and ensure accountability.
- ▶ Cross-sector collaboration to harmonize standards, share best practices, and accelerate progress (European Commission, EU Chips Act and Green Deal Initiatives, 2023).

Boards must move from observers of technological change to architects of strategic, responsible innovation. The enterprises that rise to this imperative will not only endure—they will define the next chapter of leadership in the AI era.



Conclusion: A New Standard for AI Leadership

The future of AI will not only be determined by speed of innovation but by quality of leadership. Sustainability will be a defining characteristic of enterprise credibility, resilience, and value creation.

Boards that engage with this challenge early — framing AI through the lenses of environmental stewardship, operational excellence, and societal trust — will not merely safeguard enterprise longevity. They will set the standards for responsible leadership in the AI-driven global economy.

The Next Imperative is clear:

Embedding sustainability at scale is not a constraint on innovation. It is the foundation of enduring leadership.

Closing The Next Imperative Series

This article concludes The Next Imperative series — a strategic examination of the forces reshaping AI leadership in a rapidly evolving world. Throughout four chapters, we have framed an integrated board-level agenda:

- ▶ **The Next Imperative: AI's Strategic Awakening**
Reframing AI governance as transformational foresight.
- ▶ **Capacity: From Model to Megawatt**
Recognizing compute, energy, and infrastructure as strategic assets.
- ▶ **Geopolitics: The New Chessboard for AI Sovereignty**
Understanding the new dynamics of influence, resilience, and technological sovereignty.
- ▶ **Sustainability at Scale: Rethinking AI's Energy, Ethics, and Governance**
Aligning innovation with environmental and societal stewardship.



Across each theme, the call remains consistent: **Boards must move from observers of technological change to architects of strategic, responsible innovation.** The enterprises that rise to this imperative will not only endure. They will define the next chapter of leadership in the AI era.

The Next AI Imperative: Capacity, Geopolitics, and Sustainability



Capacity: From Model to Megawatt - Rethinking AI Leadership Through Infrastructure



The Next Imperative: Geopolitics and the AI Power Play



The Next Imperative: Sustainability at Scale — Rethinking AI's Energy, Ethics, and Governance



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The Next AI Imperative: Ethics, Sovereignty, and Cyber-Resilient Systems

A strategic blueprint for AI's future at the intersection of defense, diplomacy, and disruption.



Andreea Bulisache



Fumbi Chima

A Strategic Compass for Defense, Diplomacy, and Disruption

A strategic blueprint for AI's future at the intersection of defense, diplomacy, and disruption.

The Rising Fault Lines

Artificial Intelligence has moved beyond experimentation. In 2025, it is now deeply embedded in the arteries of global commerce, national defense, financial systems, and daily life. Yet with this integration comes an uncomfortable truth: AI is not only a tool of progress but also a vector of vulnerability.

Boards and governments alike face a dual imperative to accelerate AI adoption for competitive advantage while safeguarding against risks that span **cybersecurity, ethics, and sovereignty**. This tension is no longer abstract. From **deepfake-enabled disinformation campaigns during elections** (Reuters, 2025) to **AI-driven zero-day exploits targeting critical infrastructure** (WEF, 2025), the reality is pressing: we are entering an age where power lies not only in AI capacity, but in how securely and responsibly it is governed.

This article builds on earlier imperatives — **Capacity, Geopolitics, and Sustainability** to outline the next frontier: how boards, investors, and policymakers must align **ethics, security, and sovereignty** to shape AI's role as an amplifier of human agency, not a multiplier of systemic fragility.

Cybersecurity at the Core: From Perimeter to Cognitive Defense

Traditional cybersecurity frameworks were built around **networks and endpoints**. AI changes the calculus. Today's attacks exploit **the model itself, poisoning** data, manipulating outputs, and hijacking decision loops.

According to the **World Economic Forum's Global Cybersecurity Outlook 2025**, over **60% of cybersecurity leaders now classify AI-enabled attacks as the most significant emerging threat**, surpassing ransomware and insider risks. The rise of **autonomous cyber agents** that can probe, learn, and exploit weaknesses at machine speed marks a paradigm shift: defenders no longer face human adversaries, but adaptive AI rivals.



Examples abound:

- **Healthcare systems** in Asia saw patient records compromised by AI-crafted phishing indistinguishable from legitimate medical portals.
- **Energy grids in Europe** reported attempted intrusions where generative AI synthesized operator commands to bypass human verification.
- **Financial institutions in the US** are contending with algorithmic manipulation of trading signals, blurring the line between market volatility and hostile interference.

Boards must recognize cybersecurity not as a compliance line-item, but as a strategic enabler of trust in AI adoption.

Boards should be asking:

01

How resilient are our AI models against adversarial manipulation and data poisoning?

02

Are we investing in offensive simulation capabilities “red teaming” with AI to test and harden defenses?

03

Do we have cyber crisis protocols that assume autonomous, adaptive adversaries rather than static threats?



Ethics as a Strategic Asset: Beyond Compliance to Competitive Advantage

While regulation from the **EU AI Act (2024)** to the **US AI Executive Order (2023, updated 2025)** has sharpened the focus on ethical AI, most organizations still approach it as a defensive necessity. Yet **ethical differentiation is becoming a source of trust and market advantage.**

Consider:

- **Microsoft’s AI principles** increasingly shape enterprise procurement decisions, with boards demanding ethical guarantees as part of RFPs.
- **Singapore’s Model AI Governance Framework (2024 update)** is now cited as a blueprint by emerging markets, showing how regulatory foresight can build investor confidence.
- **Consumer surveys in 2025 (Pew, Edelman Trust Barometer)** highlight a growing willingness to switch brands based on AI practices, particularly around bias and transparency.

The ethical stakes are also geopolitical. Nations exporting AI systems encoded with different value-sets, whether surveillance-heavy, profit-maximizing, or rights-preserving are not just competing on tech, but on **moral infrastructure**.

For boards, ethics is not “soft governance” but a **strategic shield** against reputational collapse and regulatory exclusion.

Boards should be asking:

- How are we embedding ethical guardrails into AI development and deployment beyond mere compliance?
- Could our AI practices become a competitive differentiator in winning customers, talent, and investors?
- Do we have visibility into the value-frameworks embedded in third-party AI tools we procure?

Sovereignty in the Age of AI: Who Owns the Future?

The AI race is as much about **control of supply chains and data flows** as it is about algorithms. Sovereignty is no longer about physical territory but about **digital autonomy**.

- **China's chip strategy** continues to prioritize self-sufficiency, even as US-led export controls tighten in 2025 (CSIS, 2025).
- **The EU's Gaia-X initiative** has gained traction, signaling Europe's commitment to “data sovereignty” and reduced dependency on US hyperscalers.
- **India's Digital Personal Data Protection Act (2023, operationalized in 2025)** demonstrates how emerging economies are framing AI sovereignty as part of national security.

For corporations, sovereignty translates into **supply chain resilience** and **regulatory alignment**. Boards must weigh dependencies: cloud providers, GPU supply, open-source foundations, and cross-border data transfer regimes.

As **AI becomes infrastructure**, sovereignty is no longer the concern of states alone. Investors and boards are stakeholders in ensuring AI futures are not hostage to geopolitics.



Boards should be asking:

- Where are our AI dependencies chips, cloud, data concentrated, and how vulnerable are they to geopolitical shocks?
- Are we actively diversifying digital supply chains, or passively assuming continuity?
- How does our AI strategy align with or expose us to national sovereignty agendas in the markets we operate in?

Towards Convergence: Building the AI Governance Loop

Capacity, Geopolitics, Sustainability and now Cybersecurity. These imperatives form an interconnected loop. AI's future cannot be governed through isolated silos; it requires **integrated governance architectures and or frameworks**.

Forward-looking boards and governments are experimenting with new models:

Cross-functional AI risk committees bringing together cybersecurity, ethics, and strategy at the board level.

"AI NATO" proposals (Brookings, 2025) calling for allied democracies to coordinate AI defense postures.

Private equity increasingly is tying investment decisions to AI governance maturity, recognizing that cyber/ethical lapses can collapse enterprise value overnight.

The challenge and opportunity is to shift from **reactive oversight** to **strategic orchestration**. In doing so, organizations can position AI as not just efficient, but enduring, trusted, and geopolitically resilient.

Boards should be asking:

- Do we treat AI governance as a fragmented compliance checklist, or as a coherent loop connecting capacity, ethics, security, and sovereignty?
- Are we engaging with peers, regulators, and even competitors to shape collective norms, rather than waiting to be shaped by them?
- What bold moves — in the next quarter — could reinforce trust, resilience, and leadership in AI governance?

Conclusion: The Imperative of Choice

AI's future won't be uniform. It will be a battleground of ideas, values, and power. But its trajectory depends on the choices of today's leaders. Boards have a once-in-a-generation opportunity to shape AI as a **human amplifier**: resilient, ethical, sovereign rather than a system burdened by unintended consequences.

As earlier articles in this series argued:

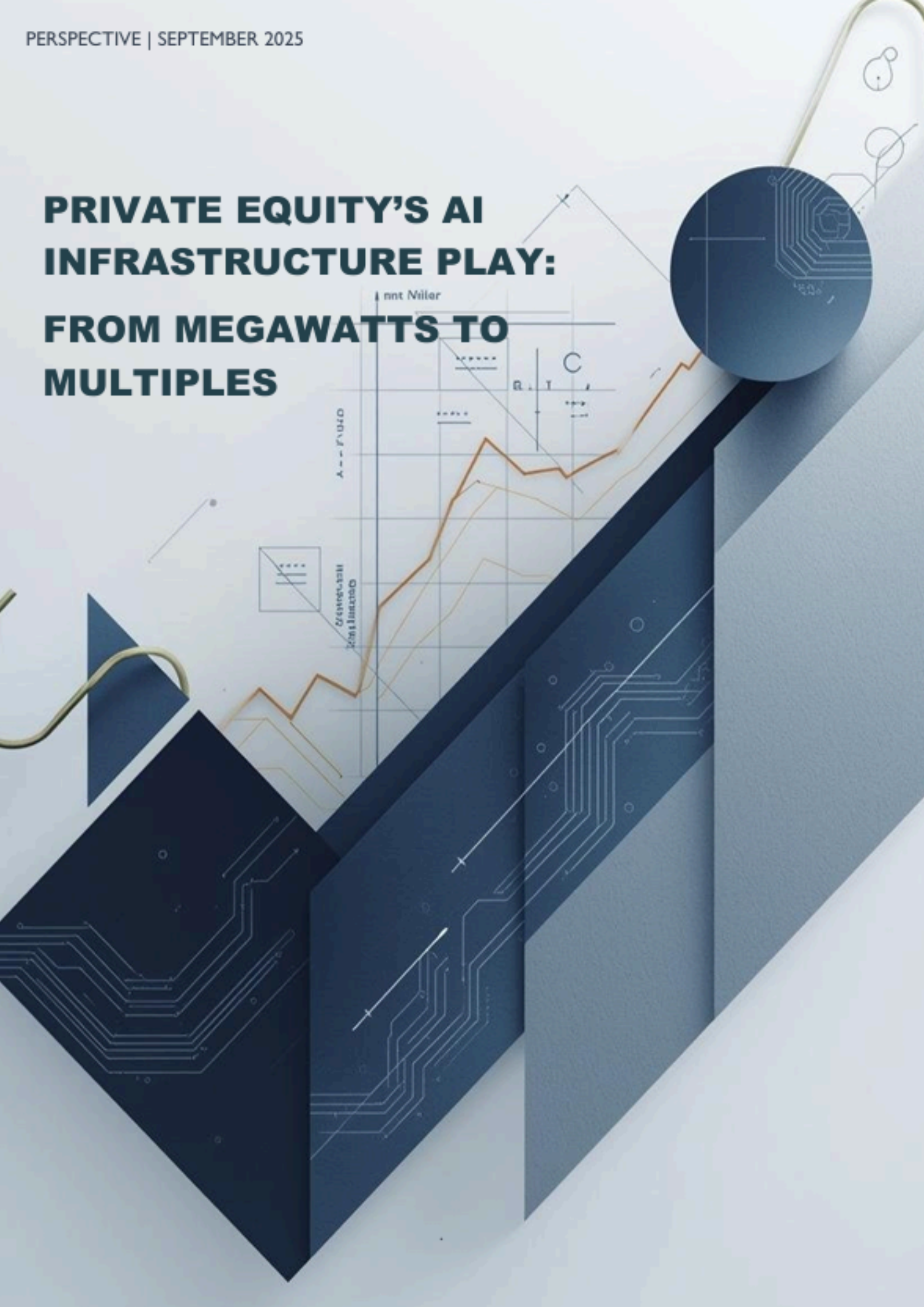
- **Capacity** defines how far AI can scale.
- **Geopolitics** defines where and under whose rules it scales.
- **Sustainability** defines whether it scales responsibly.
- **Cybersecurity and Sovereignty** now define whether it scales safely at all.

The next imperative is clear: align ethics, security, and sovereignty into a blueprint for resilience. Because in an AI-driven world, trust is the ultimate competitive advantage.

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PRIVATE EQUITY'S AI INFRASTRUCTURE PLAY: FROM MEGAWATTS TO MULTIPLES



Introduction: Why Now

Artificial intelligence has moved from the realm of speculation to an undeniable economic driver. What was once discussed in terms of algorithms and model accuracy is now constrained by something much more fundamental: **the infrastructure needed to power it**. Compute capacity, data centers, energy supply, and cooling are no longer technical details; they are macroeconomic bottlenecks.

Over the past year, we've argued that AI is not just a technology trend, it is a board-level governance challenge. In *The Next AI Imperative: Capacity* (May 2025), we highlighted that compute and energy availability, not algorithms, will define competitive advantage. In *Geopolitics and the AI Power Play* (May 2025), we showed how AI is now a sovereign-grade asset, subject to export controls, regulatory fragmentation, and supply chain volatility. And in *Capacity, Geopolitics, and Sustainability* (April 2025), we brought these threads together, pointing to the **triple challenge boards must confront: securing capacity, navigating geopolitical fractures, and aligning with sustainability demands**.

Private equity sits at the intersection of all three imperatives. The capital-intensive build-out of AI infrastructure — hyperscale data centers, energy partnerships, chip supply, and cooling innovations is increasingly being financed, structured, and governed by PE.

What's more, PE has the distinctive ability not just to build capacity but to accelerate adoption across portfolio companies, turning infrastructure into operational leverage. The numbers tell the story:

- In the U.S., DigitalBridge, Silver Lake, and Vantage are financing a \$25 billion AI data center campus in Texas, with nearly \$22 billion of debt underwritten on a scale usually reserved for sovereign projects.
- In Europe, Brookfield has committed €20 billion to French AI infrastructure via its Data4 platform, even as local regulators in Ireland and the Netherlands impose moratoria on new builds.
- In Asia-Pacific, Macquarie-backed AirTrunk refinanced A\$16 billion in sustainability-linked loans to expand its footprint in Singapore, Japan, and Australia, while KKR pursues a \$5 billion stake in Singapore-based STT GDC.
- In the Middle East, sovereign wealth funds like ADIA, Mubadala, and PIF are co-financing AI-ready campuses powered by solar, hydrogen, and nuclear, positioning themselves as energy-rich hubs for compute.

This global wave creates extraordinary opportunities but also extraordinary risk. Assets can just as easily become stranded if boards fail to anticipate power shortages, regulatory shifts, or technological obsolescence. The challenge for PE boards is not whether to invest, but how to **turn megawatts into multiples** through disciplined underwriting, resilient governance, and portfolio-wide value creation.

Underwriting the Global AI Infrastructure Bet

Traditional diligence frameworks fall short when applied to AI infrastructure. Directors must now interrogate not just lease-up assumptions but the full stack of risks: revenue resilience, energy access, regulation, financing, and technology curve dynamics.

1. Revenue Resilience

Leasing remains the lifeblood of data centers, but the profile is shifting. In North America, hyperscalers like Microsoft and Google often pre-lease entire facilities. This provides immediate revenue certainty yet creates tenant concentration risk, where the exit depends on a small handful of global buyers. Europe presents a different picture: shorter leases, stricter ESG disclosure obligations, and cautious pre-commitments due to regulatory scrutiny. Asia-Pacific is more complex still, with demand surging in Tokyo, Mumbai, and Singapore but constrained by land scarcity and localization laws.

Board lens:

- What percentage of revenue is pre-committed versus speculative?
- How concentrated is the tenant base, and how resilient is cash flow if one hyperscaler renegotiates?
- Do exit options extend beyond hyperscalers such as REITs, infra funds, or sovereign wealth buyers?

2. Power as Destiny

Energy availability has become the decisive factor in underwriting. Rack densities now exceed 250kW, and training large AI models consumes as much energy as small cities. Without guaranteed access to reliable, carbon-conscious power, projects risk failing before they start.

In the U.S., Blackstone's QTS has pioneered long-term partnerships with utilities in Iowa and Pennsylvania to lock in decades of supply. Europe, by contrast, faces grid congestion and outright political restrictions; Dublin has paused new connections until 2028, and Amsterdam has capped hyperscale growth. Asia-Pacific shows both opportunity and constraint: Singapore has cautiously reopened its pause on new data centers but ties permits to strict sustainability quotas, while Australia offers abundant renewable energy for export-oriented campuses. In the Middle East, AI data centers are being co-sited with massive solar or nuclear projects, branding themselves as "carbon-neutral compute hubs."

Board lens:

- Are we securing long-term power purchase agreements (PPAs) or virtual PPAs (VPPAs) that hedge against price volatility?
- How exposed are our assets to grid congestion or political moratoria?
- Do we have a fallback plan? such as onsite generation or utility joint ventures — if primary supply fails?

3. Regulatory and Geopolitical Headwinds

AI infrastructure is now entangled in geopolitics. Data centers are no longer viewed simply as commercial real estate but as **strategic national assets**.

In the U.S., federal incentives encourage expansion, but states like Oregon and Arizona scrutinize water consumption, limiting new approvals. The EU enforces dual compliance: carbon disclosures under the Green Deal and forthcoming AI Act obligations. In Asia, India requires strict data localization, keeping workloads within national borders, while China ties approvals directly to state-aligned AI priorities. Singapore's permitting regime awards scarce slots to projects with the strongest sustainability credentials.

Board lens:

- Which jurisdictions carry latent regulatory or political risks that could derail timelines?
- Have compliance and disclosure costs been properly embedded into IRR assumptions?
- How do export controls on chips, particularly U.S.–China tensions, affect demand forecasts and site viability?

4. Financing Innovation

The capital required to build AI infrastructure dwarfs traditional project finance. The response has been a wave of financing innovation.

AirTrunk's \$16 billion sustainability-linked refinancing set a new record, tying financing terms to environmental performance. In Texas, debt underwrites of \$22 billion approach sovereign scale, demonstrating investor appetite for AI infrastructure as a new asset class. Meanwhile, private credit arms like Blackstone Credit are stepping in to provide billion-dollar facilities, blurring the lines between infrastructure, private equity, and private credit.

Board lens:

- Do our capital structures include flexibility such as refinancing triggers or continuation vehicles if market conditions shift?
- Are sustainability-linked covenants structured as value enhancers, or do they risk constraining operational flexibility?
- How diversified is our investor base across infra funds, sovereigns, private credit, and strategic partners?

5. Technology Curve Risk

Perhaps the most under-appreciated risk lies in the technology curve. Demand forecasts often assume linear growth in GPU consumption. Yet breakthroughs in efficiency, model compression, or cooling technology could radically shift utilization.

Boards must recognize that today's capex decisions may lock investors into infrastructure that becomes obsolete. For example, liquid immersion cooling and direct-to-chip cooling are racing to become standards but the wrong bet could mean billions in stranded assets. Similarly, inference efficiency gains may reduce GPU intensity, flattening demand curves that currently appear exponential.

Board lens:

- What is the stranded-asset risk if rack densities double within 24 months?
- Are depreciation schedules realistic, or are they lagging behind technology change?
- Have downside efficiency scenarios been stress-tested as rigorously as growth cases?

Beyond Bricks: Portfolio Acceleration as Alpha

Private equity's unique advantage lies not only in financing infrastructure but in using it to **accelerate portfolio performance**. Compute capacity can become a moat across an entire fund, not just a single asset. Shared GPU blocks allow funds to allocate compute across multiple portfolio companies, lowering costs while differentiating their holdings. Centralized AI Centers of Excellence create standardized playbooks for adoption, ensuring that AI is deployed consistently and responsibly across operating companies. And by embedding AI into pricing, supply chains, and working capital management, funds can unlock **outsized returns**. **Bain estimates 2x faster adoption with centralized COEs, while McKinsey projects up to 10x ROI across portfolios.**

Board lens:

- Do we have a repeatable, fund-level AI adoption strategy?
- How are we measuring AI ROI across portfolio companies, and are those metrics board-visible?
- Can compute access itself become a shared utility and competitive advantage across our holdings?

Governance: The Global Board Imperative

AI infrastructure cannot be governed as a traditional asset class. Boards must recognize it as simultaneously **operational, geopolitical, and ethical**.

That means oversight must stretch beyond financial performance into areas like:

- **Energy Security:** Redundancy across grids, jurisdictions, and energy sources.
- **Geo-Regulatory Risk:** Exposure to localization rules, export controls, and AI regulations.
- **Sustainability:** Transparent disclosures on carbon intensity, water usage, and electronic waste.
- **Scenario Planning:** Stress-testing both exponential GPU growth and efficiency-driven contraction.
- **Exit Optionality:** Understanding the buyer universe, the hyperscalers, infra funds, sovereigns, REITs and aligning governance to preserve strategic value at exit.

The Boardroom Toolkit

To operationalize this, boards should insist on:

- **Supply chain maps** of chips, cooling technology, and energy sources.
- **ESG dashboards** aligned with LP scrutiny and regulatory standards.
- **Scenario matrices** that model moratoria, energy shocks, and efficiency breakthroughs.
- **Portfolio AI adoption scorecards** to track value creation across holdings.
- **Exit playbooks** with clear strategies for different buyer classes and geographies.

Conclusion: Stewardship at Scale

The AI era's winners will not be defined only by who builds the best models. They will be defined by those who **govern the infrastructure that powers them**.

For private equity, this is both opportunity and responsibility. Capital is being deployed at a sovereign scale, but without disciplined governance of power, regulation, sustainability, and technology risk; those megawatts may never become multiples.

As advisors and board members who have written extensively on *The Next AI Imperative* and co-led conversations on AI's governance, we bring a tested framework: one that integrates capacity, geopolitics, sustainability, and portfolio acceleration into board oversight.

For PE boards and ICs, the imperative is clear: steward AI infrastructure with foresight, discipline, and resilience. The task is not simply to finance the future, but to govern it.

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